



SD-1.50 – Wheel and Axle

The dual diameter wheel has an axle supported on simple pivots in a sturdy wall mounting bracket. Each wheel has a cord wrapped around its periphery.

PRODUCT CODE: SD-1.50

Categories: Educational, SD-1.50 Wheel And Axle, SV - Theory Of Machines, SV Series - Statics & Vibrations

Tags: Efficiency , Effort , Effort Of Friction , Mechanical Advantage , SD Theory Of Machines , SD-1.50 , Simple Machines , Velocity Ratio , Wall Mounted , Wheel And Axle

DESCRIPTION

The dual diameter wheel has an axle supported on simple pivots in a sturdy wall mounting bracket. Each wheel has a cord wrapped around its periphery. The cords can be wrapped in either direction around each wheel. A load hanger is added to each cord allowing loading of the wheels using the calibrated weights set provided.

TECHNICAL SPECIFICATIONS

- Large wheel diameter: Ø150mm
- Small wheel diameter: Ø75mm
- Maximum recommended load: 200N
- Weights set: 1 x 0.1N, 2 x 0.2N, 1 x 0.5N, 1 x 1N, 2 x 2N, 2 x 5N, 2 x 10N, 2 x 20N

FEATURES & BENEFITS

- Wall mounted apparatus
- Can be mounted to SV100 Bench Mounted Frame
- Introduction to mechanics of wheel and axle
- Effort and Efficiency investigated
- One assembly; two wheels; two diameters
- Full set of weights and hangers supplied

- Experimental determination of velocity ratio and comparison with calculated value
- Determination of variation with load of effort
- Determination of variation with load of efficiency

Ordering specification

- 1 x SD-1.50 assembled
- 1 x Load hanger
- 1 x 0.1N, 2 x 0.2N, 1 x 0.5N, 1 x 1N weights
- 2 x 2N, 2 x 5N, 2 x 10N, 2 x 20N weights
- 1 x Tape measure
- Spare cord
- Instruction manual
- Test sheet

Series includes:

Theory of Machines

- SD-1.50 Wheel and Axle
- SD-1.51 Wheel and Differential Axle Apparatus
- SD-1.52 Efficiency of Screw Threads
- SD-1.53 Gear Tooth Form Apparatus
- SD-1.54 Cam and Follower Apparatus
- SD-1.55 Simple Flywheel
- SD-1.56 Epicyclic Gear Train
- SD-1.57 Castor, Camber, and King Pin Inclination
- SD-1.58 Relation Between Angular and Linear Speeds
- SD-1.59 Disc Brake Apparatus
- SD-1.60 Static & Dynamic Balancing Apparatus
- SD-1.61 Governor Apparatus
- SD-1.62 Gear Assembly Unit Combined Drives
- SD-1.63 Single Epicyclic Gear Train

Essential Accessories

- SV100: Bench Mounted Frame

Requirements

- SV100: Bench Mounted Frame
- Recommended suspension height not to exceed 1.5metres

Shipping Specifications



Volume: 0.02m³

Gross Weight: 5kg

Dimensions

Length: 0.17m

Width: 0.11m

Height: 0.15m

Order Codes

- SD-1.50 Wheel and Axle
- SV100 Bench Mounted Frame

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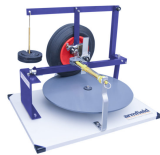
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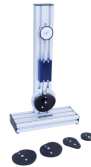
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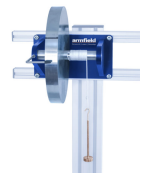
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